SFT V11.13 Gravity Addendum

Settlement Geometry, One-Speed Closure, and Root-Resolution Normalization

Companion Gravity Scaffold to Solvency Field Theory V11

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Status: Initial gravity scaffold draft for PDF review
Scope: Gate 01 through Gate 10
Use: Standalone gravity addendum / repro-synthesis note for the V11 archive

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GATE01_SCALAR_SETTLEMENT_PASS
GATE02_TENSOR_SETTLEMENT_CHANNELS_PASS
GATE03_ONE_SPEED_RECIPROCAL_CLOSURE_PASS
GATE04_PPN_WEAK_OBSERVABLE_EXTRACTION_PASS
GATE05_DIRECT_DYNAMICS_PASS
GATE06_ROBUSTNESS_AND_NORMALIZATION_TARGET_PASS
GATE07_ROOT_RESOLUTION_BRIDGE_STRUCTURAL_PASS
GATE08_ROOT_STATE_STRESS_TEST_PASS
GATE09_ROOT_INVARIANT_EQUIVALENCE_PASS
GATE10_NO_CHEAT_SCALE_SELECTION_PASS
```

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WEAK_FIELD_GRAVITY_SKELETON_EARNED
ABSOLUTE_G_SCALE_DERIVATION_OPEN
ROOT_INVARIANT_DERIVATION_OPEN
NO_PLANCK_AREA_SMUGGLING
```

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Contents

Abstract

Solvency Field Theory (SFT) seeks to describe gravity not as an independently inserted interaction, but as the large-scale settlement geometry produced by maintained real structures obeying a universal causal budget. In earlier SFT language, mass was often described as "Compton ticking." In the current formulation, that phrase is sharpened: a massive particle is treated as a real maintained one-speed internal structure whose phase, closure, and boundary write demand are different projections of one underlying process. Gravity, in this picture, is not added to the ledger; it is what the ledger's settlement geometry looks like to embedded particles.

This addendum summarizes the first V11.13 gravity-gate ladder. Gate 01 shows that a scalar settlement field sourced by maintained demand produces an exterior inverse-radius lag field, inverse-square attraction, clock-rate suppression, and the expected uniform-sphere interior scaling. Gate 02 embeds the scalar field into tensor settlement channels, preserving the scalar gravity/time-dilation projection while opening distinct mixed and spatial channels. Gate 03 tests one-speed closure and finds that reciprocal spatial closure, $B = F^{-1}$, selects the weak-field spatial response $\gamma \approx 1$. Gate 04 extracts PPN-style coefficients from the same closure and recovers $\gamma \approx 1$, $\beta \approx 1$, a light/Shapiro factor near 2, and a perihelion structure near the general-relativistic weak-field value. Gate 05 directly integrates null and timelike paths through the closure geometry and reproduces weak-field light bending and perihelion behavior. Gate 06 performs a robustness sweep and shows that the reciprocal one-speed closure exponent $\alpha = 1$ is uniquely selected among tested alternatives by GR-like weak-field dynamics.

Gate 07 then reduces the gravitational normalization problem to a root-resolution area per radian, a_ϕ, giving

$$G_{\text{eff}} = \frac{a_\phi c^3}{\hbar}. \tag{1}$$

Gate 08 stress-tests possible routes to a_ϕ and shows that c, $\hbar$, phase closure, and Compton closure alone cannot produce a universal area; all non-circular routes require one root-state dimensional primitive. Gate 09 verifies that the root area, root length, root time, root density, root tension, gravitational coupling, and full-cycle write area form one closed equivalence class. Gate 10 then performs the no-cheat scale-selection test: scale-free root-state ingredients do not select the numerical value of the root invariant, and a continuous rescaling $a_\phi \to k a_\phi$ preserves internal closure while rescaling G_{eff} by k.

The result is not a full derivation of G. Rather, the gravity problem is sharpened: SFT earns the weak-field gravity skeleton from settlement demand plus one-speed closure, while the remaining open is the derivation of the single root invariant $\mathcal{R}_{\text{root}}$ from unresolved root-state mechanics. Empirically, $a_\phi = \hbar G/c^3$, equal to the conventional Planck area, but in SFT this is interpreted only as the measured root-resolution area per radian of real phase, not as an assumed gravitational primitive.

Reader's Guide

This document is intentionally a gravity scaffold addendum. It freezes the Gate 01–10 computational and conceptual ladder so that later root-state work does not blur the gravity result.

The recommended use is:

1. Keep this note focused on weak-field gravity recovery and normalization discipline.
2. Treat Gates 01–06 as the weak-field gravity-structure scaffold.
3. Treat Gates 07–09 as the root-resolution normalization bridge and invariant compression.
4. Treat Gate 10 as the boundary marker: the gravity ladder does not by itself derive the observed

numerical value of G.

5. Move the deeper question, "why this root invariant rather than $k\mathcal{R}_{\rm root}$?", to a separate root-state / early-universe document.

The strongest claim in this document is not that SFT has completed quantum gravity or derived Einstein gravity from first principles. The strongest claim is that the tested weak-field gravity skeleton is compatible with SFT settlement language, that the one-speed reciprocal closure selects the GR-like weak-field response, and that the remaining normalization open is explicit rather than hidden.

One sentence version. Gravity is the coarse-grained settlement geometry of maintained one-speed structures; the weak-field skeleton is recovered, while the absolute coupling is reduced to one unresolved root-state invariant.

1 Purpose

The central question of this addendum is:

$$\text{Can SFT earn gravity rather than insert it?} \tag{2}$$

In standard presentations, gravity is normally introduced through a metric, curvature, an Einstein-Hilbert action, Newton's constant G, or a Newtonian potential. SFT's goal is different. The theory asks whether structures usually called gravitational can emerge from deeper ingredients:

1. real maintained structures carrying internal one-speed phase closure;
2. settlement demand generated by the maintenance of those structures;
3. a universal one-speed closure condition tying clock-rate loss, spatial path accounting, and trajectory dynamics together.

The old SFT scalar language was useful but incomplete. It often said that particles "tick at their Compton frequency." That remains a useful projection, but it is not the full picture. The current formulation treats a massive particle as a real maintained internal structure: a localized process preserving a one-speed causal budget through internal phase closure. The Compton rate is then not a little mechanical tick floating in empty space. It is the scalar clock readout of a deeper maintained geometry.

The older scalar picture was:

$$\text{mass concentrates ticking} \to \text{lag gradient} \to \text{gravity.} \tag{3}$$

The updated picture is:

$$\begin{aligned}\text{one-speed structures} &\to \text{settlement demand} \to \text{settlement geometry}\\ &\to \text{gravity/time dilation/spatial closure.}\end{aligned} \tag{4}$$

This document records the tests that move from the second line to a weak-field gravity scaffold.

2 Core SFT Gravity Dictionary

2.1 Settlement demand

A maintained real structure must continue to preserve itself as history advances. In SFT language, this maintenance generates demand on the boundary ledger. The unresolved or delayed portion of that demand is represented as a settlement or lag field.

The first scalar proxy is denoted

$$\lambda(r), \tag{5}$$

or, in tensor-promoted notation,

$$\Lambda_{00}(r). \tag{6}$$

The scalar field is not meant to be the whole gravitational field. It is the weak-field time-time/Newtonian projection of a deeper settlement structure.

2.2 Clock factor

A local settlement field suppresses clock rate by a factor

$$F(r) = \exp[-\chi\Lambda_{00}(r)], \tag{7}$$

where χ is a dimensionless gate-test coupling that sets the amplitude of the settlement potential. It is not Newton's constant.

For weak fields define

$$\phi(r) = \chi\Lambda_{00}(r), \tag{8}$$

so that

$$F(r) = e^{-\phi(r)} \approx 1 - \phi(r). \tag{9}$$

2.3 One-speed reciprocal spatial closure

The central one-speed closure proposal is that clock-rate loss and spatial path expansion are reciprocal faces of the same causal budget:

$$B(r) = F(r)^{-1} = e^{\phi(r)}. \tag{10}$$

More generally, Gates 03–06 test the family

$$B(r) = F(r)^{-\alpha} = e^{\alpha\phi(r)}. \tag{11}$$

The one-speed reciprocal closure corresponds to

$$\alpha = 1. \tag{12}$$

2.4 Metric proxy

The gates use the diagnostic weak-field metric proxy

$$ds^2 = -F(r)^2c^2dt^2 + B(r)^2d\ell^2. \tag{13}$$

For $\alpha = 1$ this becomes

$$ds^2 = -e^{-2\phi}c^2dt^2 + e^{2\phi}d\ell^2. \tag{14}$$

This proxy is not inserted as a Schwarzschild solution. It is a diagnostic representation of the SFT clock factor and spatial closure factor. The question is whether this earned closure structure reproduces weak-field gravitational behavior.

3 Gate 01 - Scalar Settlement

3.1 Question

Can maintained demand produce a scalar settlement field that yields attraction and clock-rate lag from the same object?

3.2 Setup

The scalar settlement field is solved from isotropic 3D spreading:

$$\nabla \cdot J = \sigma, \tag{15}$$

with

$$J_r = -\frac{d\lambda}{dr}. \tag{16}$$

For spherical symmetry,

$$\frac{1}{r^2}\frac{d}{dr}\left(r^2\frac{d\lambda}{dr}\right) = -\sigma. \tag{17}$$

Thus

$$\frac{d\lambda}{dr} = -\frac{Q_{\text{enclosed}}(r)}{4\pi r^2}. \tag{18}$$

A finite-grid analytic exterior tail is added to impose $\lambda(\infty) = 0$.

The clock factor is

$$F = e^{-\chi\lambda}, \tag{19}$$

and the acceleration proxy is

$$a_r = -c^2\frac{d\ln F}{dr} = c^2\chi\frac{d\lambda}{dr}. \tag{20}$$

3.3 Result

For a uniform sphere, Gate 01 found:

$$\lambda(r) \sim r^{-1}, \quad |a(r)| \sim r^{-2}, \quad 1 - F(r) \sim r^{-1}, \tag{21}$$

and inside the uniform sphere,

$$|a(r)| \sim r^{+1}. \tag{22}$$

Representative results:

- λ exterior slope: target -1, observed -1.000000.
- acceleration exterior slope: target -2, observed -2.000000.
- clock lag slope: target -1, observed -0.999865.
- interior acceleration slope: target $+1$, observed 0.999099.
- same-field coherence residual: 3.870×10^{-3} for the uniform sphere and 1.916×10^{-2} for the Gaussian profile.

3.4 Gate 01 conclusion

Gate 01 passes.

> If maintained structures create settlement demand, and settlement spreads isotropically in 3D, the same scalar field produces attraction and clock-rate lag.

This does not yet derive GR or G. It earns the first scalar gravity/time-dilation projection.

4 Gate 02 - Tensor Settlement Channels

4.1 Question

Can the scalar settlement mechanism be promoted into tensor channels without breaking the scalar result?

4.2 Setup

The scalar field is promoted as

$$\lambda \to \Lambda_{\mu\nu}. \tag{23}$$

The tested source channels are

$$\Xi_{00} = \sigma, \tag{24}$$
$$\Xi_{0x} = \beta_x \sigma, \tag{25}$$
$$\Xi_{xx} = (w + \beta_x^2)\sigma, \tag{26}$$
$$\Xi_{yy} = w\sigma, \tag{27}$$
$$\Xi_{zz} = w\sigma. \tag{28}$$

The same isotropic settlement operator is applied to each channel.

Gate 02 also tests a rotating-source mixed channel. A uniform sphere with internal circulation

$$j = \rho(\Omega \times r) \tag{29}$$

sources a mixed potential

$$A_i(x) = \int \frac{j_i(x')}{4\pi|x - x'|} \, dV'. \tag{30}$$

Along the $+x$ axis the expected behavior is $|A_y| \sim r^{-2}$, with sign reversal under $\Omega \to -\Omega$.

4.3 Result

The radial tensor projection preserved Gate 01:

- Λ_{00} slope: target -1, observed -1.000000.
- acceleration slope: target -2, observed -2.000000.
- clock lag slope: target -1, observed -0.999865.
- interior acceleration slope: target $+1$, observed 0.999099.

The channel ratios matched the defined source ratios:

$$\Lambda_{0x}/\Lambda_{00} = 0.200000, \tag{31}$$

$$\Lambda_{xx}/\Lambda_{00} = 0.140000, \tag{32}$$

$$\Lambda_{yy}/\Lambda_{00} = 0.100000, \tag{33}$$

$$\Lambda_{zz}/\Lambda_{00} = 0.100000. \tag{34}$$

The rotating-source mixed channel produced zero net current to numerical precision, nonzero angular current, exterior A_y slope near -2, and sign-flip correlation 1.000000 under $\Omega \to -\Omega$.

4.4 Gate 02 conclusion

Gate 02 passes.

> The old V11 scalar gravity picture is the weak-field 00 projection of a deeper tensor settlement field.

The important result is not that linear channel ratios reproduce defined source ratios. That part is expected. The important result is that the scalar gravity/time-dilation projection survives tensor promotion and that the mixed channel can carry circulation information with the expected sign and falloff.

5 Gate 03 - One-Speed Reciprocal Closure

5.1 Question

Does SFT's one-speed budget force a spatial response reciprocal to clock-rate suppression?

5.2 Setup

Gate 03 compares closure candidates

$$B = F^{-\alpha}, \tag{35}$$

with tested values

$$\alpha = 0, \quad 0.5, \quad 1, \quad 2. \tag{36}$$

The one-speed reciprocal closure is $\alpha = 1$, so that $FB = 1$.

The weak-field spatial-response proxy is

$$\gamma_{\text{proxy}} = \frac{\ln B}{-\ln F}. \tag{37}$$

5.3 Result

The closure comparison produced:

$$\alpha = 0 \to \gamma = 0, \tag{38}$$

$$\alpha = 0.5 \to \gamma = 0.5, \tag{39}$$

$$\alpha = 1 \to \gamma = 1, \tag{40}$$

$$\alpha = 2 \to \gamma = 2. \tag{41}$$

For $\alpha = 1$:

- $\gamma_{\log} \approx 1.000000$.
- weak strain proxy ≈ 1.000177.
- spatial strain slope ≈ -1.000135.
- $\max |FB - 1| \approx 1.110 \times 10^{-16}$.
- null-delay coefficient ≈ 2.000354.

5.4 Gate 03 conclusion

Gate 03 passes.

> Clock-rate loss and spatial path expansion are reciprocal faces of one-speed closure.

This moves SFT beyond Newtonian attraction plus clock lag. It produces the spatial response needed for weak-field light-path structure.

6 Gate 04 - PPN / Weak-Observable Extraction

6.1 Question

Does the one-speed closure proxy extract the correct weak-field relativistic observable factors?

6.2 Setup

Use

$$F = e^{-\phi}, \qquad B = e^{\alpha\phi}. \tag{42}$$

The metric proxy is

$$g_{00} = -F^2 = -e^{-2\phi}, \tag{43}$$

$$g_{ij} = B^2\delta_{ij} = e^{2\alpha\phi}\delta_{ij}. \tag{44}$$

Expanding:

$$g_{00} \approx -1 + 2\phi - 2\phi^2 + \cdots, \tag{45}$$

$$g_{ij} \approx (1 + 2\alpha\phi + \cdots)\delta_{ij}. \tag{46}$$

Thus the diagnostic expectations are $\beta \approx 1$ and $\gamma \approx \alpha$.

For one-speed reciprocal closure $\alpha = 1$, the targets are

$$\gamma \approx 1, \quad \beta \approx 1, \quad 1 + \gamma \approx 2, \quad 2 + 2\gamma - \beta \approx 3. \tag{47}$$

6.3 Result

For $\alpha = 1$:

- time coefficient A: target 2, observed 2.000000.
- β: target 1, observed 0.999403.

- spatial coefficient D: target 2, observed 2.000000.
- γ: target 1, observed 1.000000.
- light/Shapiro factor $1+\gamma$: observed 2.000000.
- perihelion numerator $2+2\gamma-\beta$: observed 3.000597.
- normalized perihelion factor: observed 1.000199.

6.4 Gate 04 conclusion

Gate 04 passes.

> SFT settlement plus one-speed reciprocal closure extracts the weak-field PPN skeleton $\gamma \approx 1$, $\beta \approx 1$.

This is not yet a full proof of GR. It is a weak-field observable extraction showing that the closure structure lands on the correct relativistic factors.

7 Gate 05 - Direct Dynamics

7.1 Question

If particles and light are propagated directly through the SFT closure geometry, do the weak-field observables appear numerically?

7.2 Setup

Use exterior settlement

$$\phi(r) = \frac{\mu}{r}, \tag{48}$$

where μ is a dimensionless dynamics amplitude in the gate scripts, not inserted G.

The closure factors are

$$F = e^{-\phi}, \qquad B = F^{-\alpha} = e^{\alpha\phi}. \tag{49}$$

The diagnostic geometry is

$$ds^2 = -F^2 dt^2 + B^2 \left(dr^2 + r^2 d\theta^2\right). \tag{50}$$

For null paths, the effective optical index is

$$n(r) = \frac{B}{F} = e^{(\alpha+1)\phi}. \tag{51}$$

Light deflection is integrated directly from the optical geometry.

For massive bound orbits, conserved quantities are computed from periapsis and apoapsis turning points in the same metric proxy, and the perihelion advance is integrated directly from the radial equation.

7.3 Result

For $\mu = 0.02$, $\alpha = 1$, and impact parameters $b = 20, 40, 80$, light deflection ratios relative to the weak-field prediction were

$$1.003154, \qquad 1.001574, \qquad 1.000786. \tag{52}$$

The ratios converge toward 1 as the weak-field parameter decreases.

For the massive orbit case, the perihelion ratio relative to the $\alpha = 1$ GR-like weak-field prediction was

$$1.000794. \tag{53}$$

7.4 Gate 05 conclusion

Gate 05 passes.

> Direct null and timelike trajectories through one-speed settlement closure reproduce weak-field light bending and perihelion structure.

This is stronger than coefficient extraction. It shows that trajectories moving through the earned closure geometry produce the expected weak-field observables.

8 Gate 06 - Robustness and Normalization Target

Gate 06 has two parts. Gate 06A performs a robustness sweep. Gate 06B identifies the normalization target required for the SFT settlement mechanism to match measured gravitational strength.

8.1 Gate 06A - Robustness sweep

Gate 06A sweeps

$$\mu = (0.005, 0.010, 0.020), \tag{54}$$

$$\alpha = (0, 0.5, 1, 2), \tag{55}$$

$$b = (20, 40, 80), \tag{56}$$

and orbit cases

$$(a, e) = (100, 0.2), \quad (200, 0.2), \quad (120, 0.5). \tag{57}$$

The RMS errors relative to the GR-like target were:

$$\alpha = 0 : 0.6005518, \tag{58}$$

$$\alpha = 0.5 : 0.3003730, \tag{59}$$

$$\alpha = 1 : 0.004622, \tag{60}$$

$$\alpha = 2 : 0.5836698. \tag{61}$$

The best alpha by GR-target RMS was $\alpha = 1$. Gate 06A passes.

8.2 Gate 06B - Normalization target

Let one complete maintenance cycle write boundary area A_0. A mass M has Compton rate

$$\nu_C = \frac{Mc^2}{h}. \tag{62}$$

The boundary area-rate is

$$\dot{A} = A_0 \nu_C. \tag{63}$$

The SFT settlement length scale is

$$\mu_{\text{SFT}} = \frac{\dot{A}}{c} = A_0 \frac{Mc}{h}. \tag{64}$$

The observed gravitational length scale is

$$\mu_G = \frac{GM}{c^2}. \tag{65}$$

Equating them gives

$$A_0 \frac{Mc}{h} = \frac{GM}{c^2}. \tag{66}$$

Mass cancels:

$$A_0 = \frac{hG}{c^3}. \tag{67}$$

Using

$$\ell_P^2 = \frac{\hbar G}{c^3} = \frac{hG}{2\pi c^3}, \tag{68}$$

this is

$$A_0 = 2\pi \ell_P^2. \tag{69}$$

Gate 06B found:

$$A_0 = 1.641344122417 \times 10^{-69}\ \text{m}^2/\text{cycle}, \tag{70}$$

$$\frac{A_0}{\ell_P^2} = 6.283185307180 = 2\pi, \tag{71}$$

with mass-independence relative standard deviation near 7.0×10^{-17}.

8.3 Gate 06 conclusion

Gate 06 passes.

> The one-speed closure exponent $\alpha = 1$ is robustly selected by weak-field dynamics.

The normalization target is

$$A_0 = \frac{hG}{c^3} = 2\pi \ell_P^2. \tag{72}$$

However, this is not yet a derivation of G. It identifies the required boundary write cost.

9 Gate 07 - Root-State Resolution Bridge

9.1 Purpose

Gate 07 asks what has actually been derived and what remains open. The forbidden move is to assume ℓ_P^2 or G and call gravity derived. Since

$$\ell_P^2 = \frac{\hbar G}{c^3}, \tag{73}$$

assuming Planck area would smuggle in G.

Instead, Gate 07 introduces a root-resolution area per radian

$$a_\phi. \tag{74}$$

This area is not derived in Gate 07. It is isolated as the remaining primitive or theorem target.

9.2 Dimensional no-go

Gate 07 first verifies that c, $\hbar$, and dimensionless phase alone cannot produce a universal area. With dimensions $[c] = LT^{-1}$ and $[\hbar] = ML^2T^{-1}$, there is no combination $c^x\hbar^y$ with dimensions L^2 and no mass or time residue. The exact dimension equations contradict.

An area primitive or deeper area theorem is required.

9.3 Gate 07A - Compton closure earns 2π

Use the dimensionless closure count

$$I = \frac{2\pi ER}{\hbar c}. \tag{75}$$

For a massive mode at its reduced Compton radius,

$$E = mc^2, \qquad R_C = \frac{\hbar}{mc}. \tag{76}$$

Then

$$I = \frac{2\pi(mc^2)(\hbar/mc)}{\hbar c} = 2\pi. \tag{77}$$

Gate 07A found $I/(2\pi) = 1.000000000000000$, with maximum numerical error near 1.1×10^{-16}.

9.4 Gate 07B - Root-resolution bridge

Introduce the root-resolution area per radian:

$$dA = a_\phi\, d\phi. \tag{78}$$

A complete cycle has $\Delta\phi = 2\pi$. Therefore

$$A_0 = \int_0^{2\pi} a_\phi\, d\phi = 2\pi a_\phi. \tag{79}$$

The Gate 06 normalization relation is

$$G_{\text{eff}} = \frac{A_0 c^3}{h}. \tag{80}$$

Substituting $A_0 = 2\pi a_\phi$ and $h = 2\pi\hbar$ gives

$$G_{\text{eff}} = \frac{a_\phi c^3}{\hbar}. \tag{81}$$

For mass M,

$$\mu_{\text{SFT}} = \frac{A_0 \nu_C}{c} = \frac{a_\phi M c}{\hbar} = \frac{G_{\text{eff}} M}{c^2}. \tag{82}$$

Gate 07B passes.

9.5 Gate 07C - Empirical calibration only

Gate 07C uses measured G only after the structural bridge is complete. It identifies

$$a_\phi^{\text{empirical}} = \frac{\hbar G}{c^3} = 2.612280303975 \times 10^{-70}\ \text{m}^2. \tag{83}$$

The full-cycle area is

$$A_0 = 2\pi a_\phi = 1.641344122417 \times 10^{-69}\ \text{m}^2/\text{cycle}. \tag{84}$$

This is a calibration, not a derivation.

9.6 Gate 07 conclusion

Gate 07 is a structural pass. It earns the 2π phase/write count from Compton closure, exposes the need for an area primitive or root-area theorem, and derives the bridge

$$G_{\text{eff}} = \frac{a_\phi c^3}{\hbar}. \tag{85}$$

The remaining open is deriving a_ϕ itself from unresolved root-state mechanics.

10 Gate 08 - Root-State Stress Test

Gate 08 stress-tests possible routes to the remaining normalization primitive a_ϕ. The purpose is not to derive a_ϕ, but to prevent circular derivations and classify which routes are valid, invalid, or merely equivalent to supplying the missing root-state dimensional primitive.

10.1 Routes tested

Gate 08 classifies these candidate routes:

1. $c + \hbar$ only;
2. particle-specific Compton-area routes using m_e or m_p;
3. cosmological-scale routes using H_0;
4. root-area primitive route using a_ϕ;
5. root-density route using ρ_A^{-1};

6. root-length route using l^2_{root};
7. root-time route using $c^2 t^2_{\mathrm{root}}$;
8. root-tension route using $\hbar c/F_{\mathrm{root}}$;
9. forbidden circular route using $\hbar G/c^3$.

The circularity rule remains strict: G, ℓ_P, Planck area, Schwarzschild radius, and black-hole area entropy are not allowed as derivation inputs.

10.2 Results

The $c + \hbar$ only route fails dimensionally. Phase closure and Compton closure can earn a dimensionless 2π count, but they cannot produce a universal area.

Particle-mass routes can form areas of the type

$$\frac{\hbar^2}{m^2 c^2}, \tag{86}$$

but these are particle-specific Compton areas. Gate 08 found:

$$a_e \approx 1.491190 \times 10^{-25}\ \mathrm{m}^2, \tag{87}$$
$$a_p \approx 4.422984 \times 10^{-32}\ \mathrm{m}^2. \tag{88}$$

They are not universal and therefore cannot be the root-resolution area controlling gravitational normalization.

The cosmological route using c^2/H_0^2 can form an area, but it gives an epoch-scale area:

$$\sim 1.746411 \times 10^{52}\ \mathrm{m}^2, \tag{89}$$

not a local root-resolution patch.

The route

$$a_\phi = \frac{\hbar G}{c^3} \tag{90}$$

works numerically, but it is circular if used before calibration, because it already contains G.

The valid non-circular branches all require one independent root-state dimensional primitive. This primitive may be represented equivalently as

$$a_\phi, \quad \rho_A^{-1}, \quad l^2_{\mathrm{root}}, \quad c^2 t^2_{\mathrm{root}}, \quad \frac{\hbar c}{F_{\mathrm{root}}}. \tag{91}$$

10.3 Empirical calibration

After the non-circular classification, Gate 08 calibrates the root primitive using measured G. This is explicitly not a derivation. The calibrated values are:

$$a_\phi = 2.612280303975 \times 10^{-70}\ \mathrm{m}^2, \tag{92}$$
$$A_0 = 2\pi a_\phi = 1.641344122417 \times 10^{-69}\ \mathrm{m}^2/\mathrm{cycle}, \tag{93}$$
$$l_{\mathrm{root}} = 1.616255024424 \times 10^{-35}\ \mathrm{m}, \tag{94}$$
$$t_{\mathrm{root}} = 5.391246448314 \times 10^{-44}\ \mathrm{s}, \tag{95}$$
$$\rho_A = 3.828073114812 \times 10^{69}\ \mathrm{m}^{-2}, \tag{96}$$
$$F_{\mathrm{root}} = \frac{\hbar c}{a_\phi} = \frac{c^4}{G} = 1.210255564338 \times 10^{44}\ \mathrm{N}. \tag{97}$$

10.4 Gate 08 conclusion

Gate 08 passes as a stress test. It establishes that the remaining primitive cannot be produced from c, $\hbar$, phase closure, or Compton closure alone. It rejects particle-specific and cosmological-scale substitutions. It flags the G/Planck route as circular before calibration. It shows that all valid non-circular branches require one independent root-state dimensional primitive.

11 Gate 09 - Root-Invariant Equivalence Class

Gate 09 tests whether the calibrated quantities a_ϕ, l_{root}, t_{root}, ρ_A, F_{root}, G_{eff}, and A_0 are separate assumptions or operational faces of one root invariant.

The result is a compression win. These quantities are not independent knobs. Once any one face is supplied, the entire equivalence class is fixed.

11.1 Root faces

Using empirical calibration only, Gate 09 found:

$$a_\phi = 2.612280303975 \times 10^{-70}\ \mathrm{m}^2, \tag{98}$$
$$l_{\mathrm{root}} = 1.616255024424 \times 10^{-35}\ \mathrm{m}, \tag{99}$$
$$t_{\mathrm{root}} = 5.391246448314 \times 10^{-44}\ \mathrm{s}, \tag{100}$$
$$\rho_A = 3.828073114812 \times 10^{69}\ \mathrm{m}^{-2}, \tag{101}$$
$$F_{\mathrm{root}} = 1.210255564338 \times 10^{44}\ \mathrm{N}, \tag{102}$$
$$G_{\mathrm{eff}} = 6.674300000000 \times 10^{-11}, \tag{103}$$
$$A_0 = 1.641344122417 \times 10^{-69}\ \mathrm{m}^2/\mathrm{cycle}. \tag{104}$$

These values are calibrated from measured G; Gate 09 does not claim to derive them from first principles.

11.2 Area reconstruction

Gate 09 reconstructs the same root area from every face:

$$a_\phi = a_\phi, \tag{105}$$
$$a_\phi = l_{\text{root}}^2, \tag{106}$$
$$a_\phi = (ct_{\text{root}})^2, \tag{107}$$
$$a_\phi = \rho_A^{-1}, \tag{108}$$
$$a_\phi = \frac{\hbar c}{F_{\text{root}}}, \tag{109}$$
$$a_\phi = \frac{\hbar G_{\text{eff}}}{c^3}, \tag{110}$$
$$a_\phi = \frac{A_0}{2\pi}. \tag{111}$$

All reconstructed areas matched a_ϕ with ratio 1.000000000000000 to numerical precision.

11.3 Closure identities

Gate 09 verified the closure identities:

$$l_{\text{root}}^2 / a_\phi = 1, \tag{112}$$
$$ct_{\text{root}} / l_{\text{root}} = 1, \tag{113}$$
$$\rho_A a_\phi = 1, \tag{114}$$
$$F_{\text{root}} a_\phi / (\hbar c) = 1, \tag{115}$$
$$G_{\text{eff}} \hbar / (a_\phi c^3) = 1, \tag{116}$$
$$A_0 / (2\pi a_\phi) = 1. \tag{117}$$

The maximum identity error was about 1.110×10^{-16}.

11.4 Single-seed reconstruction

Gate 09 tests whether any one face can reconstruct the whole root-invariant class. It can. The single-seed reconstruction max errors were:

- seed a_ϕ: 0.000×10^0;
- seed l_{root}: 2.002×10^{-16};
- seed t_{root}: 2.002×10^{-16};
- seed ρ_A: 0.000×10^0;
- seed F_{root}: 0.000×10^0;
- seed G_{eff}: 0.000×10^0;
- seed A_0: 0.000×10^0.

Thus any one face determines the full equivalence class.

11.5 Independent perturbation test

Gate 09 perturbs one face independently while leaving the others fixed. If the faces were independent constants, this would be allowed. Instead, every independent perturbation breaks closure. The minimum area-ratio spread under a 1% independent perturbation was about 9.901×10^{-3}, so the perturbation test passed. This shows the root faces are not separate free constants. They are constrained projections of one object.

11.6 Gate 09 conclusion

Gate 09 passes.

> a_ϕ, l_{root}, t_{root}, ρ_A, F_{root}, G_{eff}, and A_0 form one closed root-invariant equivalence class.

They are one root invariant expressed in different operational languages: area, length, time, density, tension, gravitational coupling, and full-cycle write area. The remaining open is compressed to a single object:

$$\mathcal{R}_{\text{root}}. \tag{118}$$

12 Gate 10 - Root-State Scale Selection / No-Cheat Test

Gate 10 asks whether the single unresolved root invariant $\mathcal{R}_{\text{root}}$ can be selected from scale-free root-state ingredients without smuggling in gravitational information.

The result is an important boundary marker. Gates 01–09 earn gravity's weak-field structure, reduce the gravitational normalization to a root invariant, and show that all root faces are one equivalence class. Gate 10 shows that this is still not enough to select the observed numerical strength of gravity.

12.1 Routes tested

Gate 10 classifies the following scale-selection routes:

1. phase only;
2. $c + \hbar +$ phase;
3. $c + \hbar$ Compton closure without a mass scale;
4. electron Compton area;
5. proton Compton area;
6. cosmological horizon area using c^2/H_0^2;
7. root distinguishability density;
8. root area primitive;
9. root causal length;
10. root causal time plus c;
11. root tension plus $\hbar, c$;
12. forbidden $\hbar G/c^3$ route.

12.2 Results

The scale-free routes do not select an absolute area:

- phase only: no dimensional scale;
- $c + \hbar +$ phase: no area;
- $c + \hbar$ without a mass scale: no area.

Particle Compton-area routes can form areas, but they are particle-specific rather than universal:

$$\text{electron route} \approx 1.491189802321 \times 10^{-25}\ \text{m}^2, \tag{119}$$
$$\text{proton route} \approx 4.422983776653 \times 10^{-32}\ \text{m}^2. \tag{120}$$

The cosmological route can form an area of about $1.746410831513 \times 10^{52}$ m^2, but this is an epoch-scale horizon area, not a local inherited root-resolution primitive.

The route $a_\phi = \hbar G/c^3$ lands on the calibrated value, but it is circular before calibration because it already contains G.

Root-density, root-area, root-length, root-time, and root-tension branches are valid only if they are explicitly postulated or derived from some new non-gravitational root-state condition. Gate 10 does not derive them.

12.3 Scale degeneracy sweep

Gate 10 performs the transformation

$$a_\phi \to k a_\phi. \tag{121}$$

The internal root-invariant closure remains valid for every tested k, while the effective gravitational coupling scales as

$$G_{\text{eff}} \to k G_{\text{eff}}. \tag{122}$$

The sweep ran from $k = 10^{-6}$, giving $G_{\text{eff}}/G = 10^{-6}$, to $k = 10^6$, giving $G_{\text{eff}}/G = 10^6$. The maximum internal closure error was about 2.220×10^{-16}.

This proves that the gravity ladder's internal closure does not by itself select the observed value of G.

12.4 Gate 10 conclusion

Gate 10 passes as a no-cheat test.

> Gates 01–09 earn gravity's structure, but they do not yet derive gravity's absolute scale.

The remaining open is to derive $\mathcal{R}_{\text{root}}$ from a new non-circular root-state condition or state $\mathcal{R}_{\text{root}}$ honestly as an inherited primitive/invariant. This is not a failure of the gravity ladder. It is a clean boundary marker showing exactly where the next theory layer begins.

13 Combined Gravity Ladder

The V11.13 ladder can be summarized as:

$$\text{Gate 01} : \text{scalar settlement gives lag, attraction, and clock suppression,} \tag{123}$$
$$\text{Gate 02} : \text{scalar projection embeds in tensor settlement channels,} \tag{124}$$
$$\text{Gate 03} : \text{one-speed reciprocal closure selects } B = F^{-1}, \tag{125}$$
$$\text{Gate 04} : \text{weak PPN-style factors are extracted,} \tag{126}$$
$$\text{Gate 05} : \text{direct null/timelike dynamics reproduce weak-field observables,} \tag{127}$$
$$\text{Gate 06} : \alpha = 1 \text{ is robust and the normalization target is identified,} \tag{128}$$
$$\text{Gate 07} : G \text{ is reduced to root-resolution area per radian,} \tag{129}$$
$$\text{Gate 08} : \text{non-circularity stress test requires a root primitive,} \tag{130}$$
$$\text{Gate 09} : \text{root faces compress into one invariant,} \tag{131}$$
$$\text{Gate 10} : \text{scale-free closure does not select the invariant's value.} \tag{132}$$

The compact technical chain is:

$$\text{settlement demand} \to \Lambda_{00} \to F = e^{-\phi} \to B = F^{-1} \to \gamma \approx 1,\ \beta \approx 1. \tag{133}$$

The normalization chain is:

$$\nu_C = \frac{Mc^2}{h}, \qquad A_0 = 2\pi a_\phi, \tag{134}$$

$$\mu_{\mathrm{SFT}} = \frac{A_0 \nu_C}{c} = \frac{a_\phi M c}{\hbar} = \frac{G_{\mathrm{eff}} M}{c^2}, \tag{135}$$

so

$$G_{\mathrm{eff}} = \frac{a_\phi c^3}{\hbar}. \tag{136}$$

14 Gate Status Table

Gate	Theme	Result	Caveat
01	Scalar settlement	Pass	Scalar projection only; no G derivation
02	Tensor settlement channels	Pass	Channel operator linearity is expected; full dynamics open
03	One-speed reciprocal closure	Pass	Closure proxy, not full metric derivation
04	PPN/weak-observable extraction	Pass	Coefficient extraction, not full GR proof
05	Direct dynamics	Pass	Weak-field diagnostic geometry only
06	Robustness and normalization target	Pass	Identifies $A_0 = hG/c^3$, does not derive it
07	Root-resolution bridge	Structural pass	Derives $G_{\mathrm{eff}} = a_\phi c^3/\hbar$ if a_ϕ is supplied
08	Root-state stress test	Pass	Shows root primitive is required; does not derive it
09	Root-invariant equivalence	Pass	Compresses faces to one invariant; value still empirical

Gate	Theme	Result	Caveat
10	Scale-selection/no-cheat test	Pass	Proves scale-free closure alone does not fix observed G

15 What This Addendum Establishes

This addendum establishes the following:

1. Maintained demand with isotropic settlement produces an exterior inverse-radius scalar field.
2. The same scalar field gives inverse-square attraction and clock-rate suppression.
3. The scalar field embeds into tensor settlement channels.
4. A rotating-source mixed channel carries circulation information with the expected falloff and sign reversal.
5. One-speed reciprocal closure selects $B = F^{-1}$ and $\gamma \approx 1$ among tested closure exponents.
6. The closure proxy extracts $\gamma \approx 1$, $\beta \approx 1$, light/Shapiro factor near 2, and perihelion structure near the GR weak-field value.
7. Direct null and timelike trajectories through the closure geometry reproduce weak-field light bending and perihelion behavior.
8. Robustness sweeps select $\alpha = 1$ among tested alternatives.
9. The required normalization can be written as $A_0 = 2\pi a_\phi$ and $G_{\text{eff}} = a_\phi c^3/\hbar$.
10. Compton closure earns the 2π cycle/write count without using G.
11. c, $\hbar$, phase closure, and Compton closure alone cannot form a universal root area.
12. Particle-specific and cosmological-scale substitutes fail as universal root-resolution candidates.
13. The Planck-area route works only as empirical calibration, not as derivation.
14. The root area, root length, root time, root density, root tension, gravitational coupling, and full-cycle write area are one equivalence class.
15. A continuous rescaling $a_\phi \to k a_\phi$ preserves internal closure but rescales G_{eff} by k.
16. The observed gravitational strength therefore requires a non-circular root-state scale selector, or $\mathcal{R}_{\text{root}}$ must be stated as an inherited primitive.

16 What This Addendum Does Not Establish

This section should remain prominent. It is the guardrail that keeps the addendum scientifically honest.

This addendum does not derive the numerical value of G from first principles.

This addendum does not derive a_ϕ, l_{root}, t_{root}, ρ_A, or F_{root} from unresolved root-state mechanics.

This addendum does not prove the full Einstein field equations.

This addendum does not prove that $\Lambda_{\mu\nu}$ is the Einstein tensor.

This addendum does not claim a strong-field solution.

This addendum does not claim a black-hole entropy derivation.

This addendum does not use Planck area as a derivation input.

This addendum does not claim that the early universe was literally a tiny object embedded in pre-existing space.

This addendum does not complete quantum gravity.

This addendum establishes a scaffold: if SFT supplies maintained one-speed structures, settlement demand, reciprocal one-speed closure, and one root invariant, then the weak-field gravity skeleton and its normalization bridge are recovered in SFT ledger language. The remaining root-invariant derivation is explicit.

17 Open Items After Gate 10

17.1 Open Item 1 - Root-invariant derivation

Derive the single root invariant $\mathcal{R}_{\text{root}}$ from unresolved root-state mechanics without using G, ℓ_P, black-hole entropy, Schwarzschild radius, or Planck area as assumptions.

Target:

$$\text{root-state mechanics} \Rightarrow \mathcal{R}_{\text{root}}. \tag{137}$$

17.2 Open Item 2 - Tensor settlement dynamics

Extend the tensor settlement framework beyond the weak-field proxy. Determine whether $\Lambda_{\mu\nu}$ obeys a conservation law and whether it yields full metric dynamics.

17.3 Open Item 3 - Strong-field regime

Test whether the settlement/closure framework remains coherent near compact objects, horizons, or high-curvature regimes without importing Schwarzschild structure by hand.

17.4 Open Item 4 - Cosmological perturbations

Connect the gravity normalization and tensor settlement framework to cosmological perturbations and the existing V11/V12 sound-horizon and late-time handoff work.

17.5 Open Item 5 - Root-state companion document

Move the origin-of-resolution question into a separate document. The working title could be:

> SFT Root-State Resolution: The Origin of the First Write-Patch.

This companion would ask why the original unresolved causal/write state has this root invariant rather than any scaled value $k\mathcal{R}_{\text{root}}$.

18 Boundary of the V11.13 Gravity Note

The V11.13 gravity note should stop at the Gate 10 boundary marker. The gravity ladder can now make a strong and honest claim:

> SFT settlement demand plus one-speed closure recovers the weak-field gravity skeleton and reduces the absolute gravitational normalization to one unresolved root invariant.

It should not claim that the root invariant is derived.

The next problem is conceptually larger than weak-field gravity recovery. It is a root-state / early-universe / first-resolution problem:

> What condition in the original unresolved causal/write state fixes the first finite write-patch?

That question belongs in a separate document, not as a rushed final section of this gravity note.

19 Frozen V11.13 Gravity Claim

The frozen V11.13 gravity claim is:

`WEAK_FIELD_GRAVITY_SKELETON_EARNED`

with explicit caveats:

`ROOT_INVARIANT_DERIVATION_OPEN` `ABSOLUTE_G_SCALE_DERIVATION_OPEN`
`FULL_FIELD_EQUATIONS_OPEN` `STRONG_FIELD_TESTS_OPEN`

A more compact label is:

`GATE01_10_GRAVITY_SCAFFOLD_PASS_ROOT_SCALE_OPEN`

20 One-Line Summary

SFT settlement demand plus one-speed reciprocal closure recovers the weak-field gravity skeleton and reduces the absolute gravitational normalization to one root invariant, while Gate 10 proves that the invariant's numerical value is not selected by scale-free closure alone and must be derived in a separate root-state theory or stated honestly as an inherited primitive.

PDF Export Note

This addendum is designed to stand alone in the V11 archive.

Recommended filename:

`SFT_V11_13_Gravity_Initial_Draft.pdf`

Recommended archive placement:

```
SFT_V11_13_Gravity_Initial_Draft.pdf
SFT_V11_13_Gravity_Repro_Pack.zip
SFT_Root_State_Resolution_Companion_OPEN.pdf  (future)
```

The filename and front matter deliberately avoid implying that the root invariant, big G, full GR, or strong-field gravity has been fully derived.

A Compact Equation Chain

Compton phase rate:

$$\nu_C = \frac{mc^2}{h}. \tag{138}$$

Scalar settlement exterior:

$$\lambda(r) \sim \frac{1}{r}. \tag{139}$$

Clock factor:

$$F = e^{-\phi}, \qquad \phi = \chi\Lambda_{00}. \tag{140}$$

One-speed closure:

$$B = F^{-1} = e^{\phi}. \tag{141}$$

Metric proxy:

$$ds^2 = -F^2c^2dt^2 + B^2d\ell^2. \tag{142}$$

Weak expansion:

$$g_{00} \approx -1 + 2\phi - 2\phi^2, \qquad g_{ij} \approx (1 + 2\phi)\delta_{ij}. \tag{143}$$

Thus:

$$\gamma \approx 1, \qquad \beta \approx 1. \tag{144}$$

Root-resolution area per radian:

$$dA = a_\phi \, d\phi. \tag{145}$$

Full cycle:

$$A_0 = 2\pi a_\phi. \tag{146}$$

Gravity normalization:

$$G_{\text{eff}} = \frac{A_0c^3}{h} = \frac{a_\phi c^3}{\hbar}. \tag{147}$$

Empirical calibration:

$$a_\phi = \frac{\hbar G}{c^3}. \tag{148}$$

B Plain-English Summary

Every massive particle is a real maintained structure. It keeps itself coherent by completing internal one-speed phase closure. That maintenance creates demand on the boundary ledger. Demand that has not fully settled appears as lag. The local amount of lag slows clocks. The gradient of lag creates attraction. The one-speed rule forces a reciprocal spatial response, so light and matter move as if spacetime is curved.

The gravity gates show that this mechanism reproduces the weak-field gravity skeleton. They also show that the strength of gravity is controlled by one quantity: the boundary area cost of one radian of real internal phase. SFT calls this a_ϕ.

Measured gravity says a_ϕ equals the conventional Planck area. But SFT should not assume that. The next challenge is to explain why the unresolved root-state of the universe has that write-patch size.

In the SFT interpretation, the hot Big Bang is not the creation of a replacement universe. It is a phase transition of the same universe. The earliest phase was one unresolved causal/write state. Later particles and histories are differentiated structures that still obey the original write-resolution rule. Gravity is the macroscopic stress of preserving those differentiated structures under that inherited rule.

C Current Open Problems

1. Derive $\mathcal{R}_{\text{root}}$ from root-state mechanics without using G, ℓ_P, or black-hole entropy as assumptions.
2. Extend tensor settlement beyond weak-field proxies.
3. Determine whether settlement dynamics recover a full metric field equation.
4. Test strong-field regimes.
5. Test cosmological perturbations under the same gravity normalization.
6. Connect the root-resolution picture to the One documents and SFT phase-transition cosmology.
7. Decide whether $\mathcal{R}_{\text{root}}$ is derivable or must be stated as an inherited primitive.

D Version Notes

This is an initial V11.13 gravity draft. It freezes the Gate 01–10 results as a coherent ladder but does not attempt the future root-state derivation. The language should remain careful: Gates 01–10 earn a weak-field skeleton, a normalization bridge, an invariant compression, and a no-cheat boundary marker. They do not complete the derivation of the root invariant, full GR, strong-field gravity, or quantum gravity.